INCH-POUND

MIL-DTL-6363/8A
30 September 1997
SUPERSEDING
MIL-L-6363/8
25 September 1992

DETAIL SPECIFICATION SHEET

LAMPS, INCANDESCENT, AIRCRAFT SERVICE, SINGLE CONTACT MIDGET FLANGED BASE T-1-3/4 BULB

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-6363.

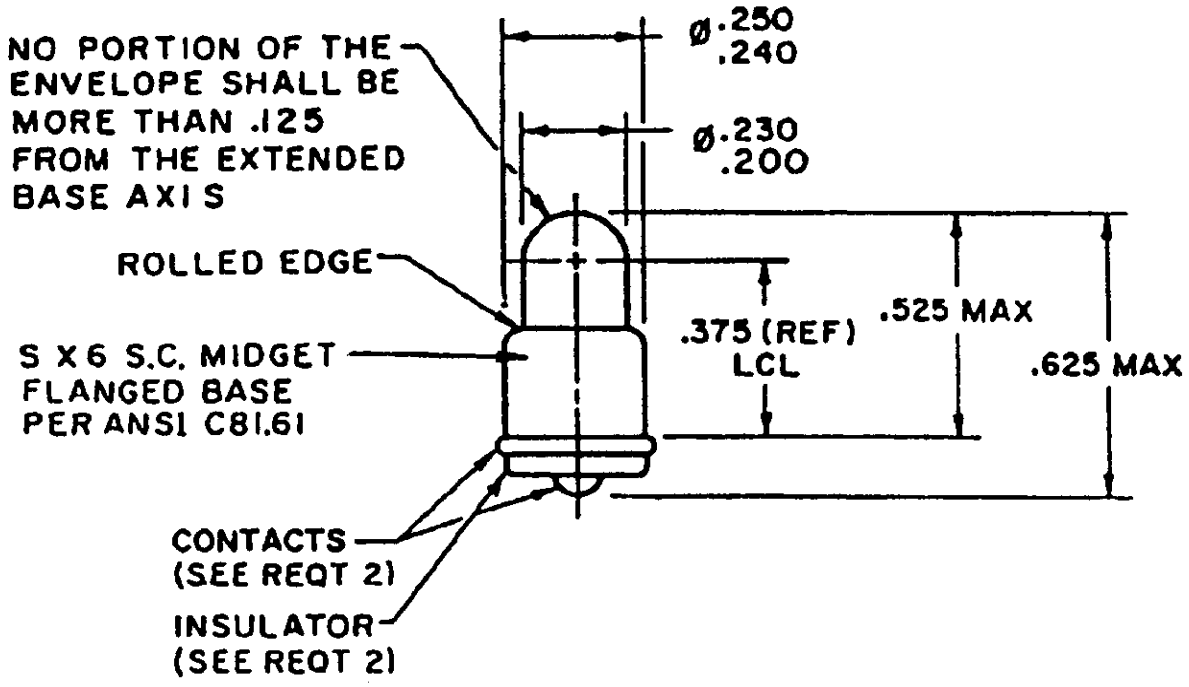


FIGURE 1. Configuration and dimensions.

See NOTES on the next page.

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NOTES:
1. Dimensions are in inches.
2. Intended use - aircraft indicator, instrument, and panel light assemblies.
3. For design feature purposes, this specification takes precedence over procurement documents referenced herein.
4. Qualification by similarity - Lamp M6363/8-7AS15 (or optional M6363/8-5 AS15) is required to be fully qualified in accordance with table I of MIL-DTL-6363. Lamp M6363/8-2 AS10 is required to be qualified in accordance with table II of MIL-DTL-6363. All other basic lamps (MSCD ±25 percent) and "AS" designated lamps may be granted qualification by similarity and extension of qualification from the qualified lamps with no additional testing. Red lamp M6363/8-7R (or optional -5R) is required to be qualified in accordance with table II of MIL-DTL-6363. All other colored lamps may be granted qualification from -7R (or optional -5R) with no additional testing. Refer to MIL-DTD-6363 for conditions and data requirements for qualification.

REQUIREMENTS:

1. Configuration and dimensions: See figure 1.

2. Construction:

   a. Standard lamp construction shall include a side weld or solder bead and solder bead bottom contact.

   b. Where required for greater reliability and added resistance to corrosion and internal shorting, solderless construction is available as an option by the addition of a dash and suffix letter "S" to the part identification number in table I.

EXAMPLE: M6363/8-1-S

When so specified, solderless construction shall include a side weld and nickel plated brass bottom contact.

   c. Insulators shall be color coded as follows to identify voltage rating of lamps:

5.0v - Green
6.0v - Green
6.3v - Green
14.0v - Yellow
28.0v - Red

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TABLE I. Part or identifying numbers (PIN) and operating characteristics.

Part or Identifying Number (PIN) (SEE REQT 2)	Type MIL-DTL-6363	Bulb Finish	Shock Test Level (G's)	Electrical Ratings			Filament Style	MSCD 1/ 2/	Average Rated Lab Life at DC
				Volts (Nom)	Amperes (±10%)	Watts (Max)		Candelas	Hours
M6363/8-1	I	Clear	50	5.0	0.06	0.33	C-2R	0.15±25%	2,000
M6363/8-1 AS15		Clear		5.0	0.06	0.33	C-2R	0.15±15%	2,000
M6363/8-2		Clear		6.0	0.20	1.32	C-2R	0.34±25%	1,000
M6363/8-2 AS10		Clear		6.0	0.20	1.32	C-2R	0.34±10%	1,000
M6363/8-2R		Red		6.0	0.20	1.32	C-2R	0.34±25%	1,000
M6363/8-2B		3/		6.0	0.20	1.32	C-2R	0.34±25%	1,000
M6363/8-3		Clear		6.3	0.20	1.38	C-2F	0.40±25%	10,000
M6363/8-3 AS15		Clear		6.3	0.20	1.38	C-2F	0.40±15%	10,000
M6363/8-4		Clear		14.0	0.10	1.54	C-2F	0.50±25%	5,000
M6363/8-4 AS15		Clear		14.0	0.10	1.54	C-2F	0.50±15%	5,000
M6363/8-5		Clear		28.0	0.04	1.24	C-2F	0.34±25%	1,500
M6363/8-5 AS15		Clear		28.0	0.04	1.24	C-2F	0.34±15%	1,500
M6363/8-5R		Red		28.0	0.04	1.24	C-2F	0.34±25%	1,500
M6363/8-6 AS15		Clear		28.0	0.04	1.24	C-2F	0.15±15%	5,000
M6363/8-7		Clear		28.0	0.04	1.24	C-2F	0.30±25%	2,000
M6363/8-7AS15		Clear		28.0	0.04	1.24	C-2F	0.30±15%	2,000
M6363/8-7R		Red		28.0	0.04	1.24	C-2F	0.30±25%	2,000

1/ MSCD for -2 and -2 AS10 is when operated at 5 volts.
2/ MSCD for colored lamps applies to clear lamp before coating is applied.
3/ Blue/White

3. Finish: Glass globe finish shall be as indicated in table I. Red lamps shall conform to MIL-C-25050 for instrument and panel lighting red. Blue/white lamps shall conform to MIL-C-25050 for blue filtered white color.

4. Lamp operating characteristics: When operating at rated voltage, the lamp operating characteristics shall be as specified in table I.

5. Where indicated by suffix "AS__" in the part number, lamps shall be aged for a minimum of ten hours at rated voltage and selected for 15 or 10 percent tolerance on MSCD as indicated in table I.

6. MSCD at 1,000 hours shall be not less than 80 percent of initial MSCD when measured at rated voltage.

7. Coating adhesion: Applicable for exterior coated colored lamps.

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8. High temperature: Applicable for qualification tests only (85°C/185F°).

9. Thermal shock: Applicable for qualification tests only. Use temperatures for Type I lamps.

10. Random vibration: Applicable for qualification tests only. See figure 2 for random vibration curve.

11. Shock: Applicable for qualification tests only. See table I for applicable “g” level.

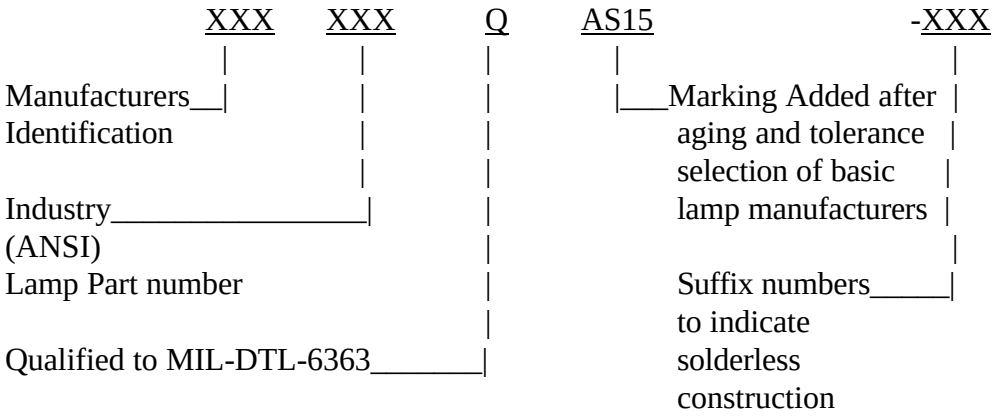
12. Humidity: Applicable for qualification tests only.

13. Salt spray: Applicable for qualification tests only.

14. Marking: Production lamps qualified to MIL-DTL-6363 and listed on QPL-6363 shall be marked in accordance with ANSI-C78.390 and as supplemented herein. ANSI-C78.390 authorized suffix letters shall be supplemented with the letter “Q” (MIL-DTL-6363 QPL item). The letter “Q” shall be incorporated into the method of designation in accordance with the preferred marking below:

EXAMPLE

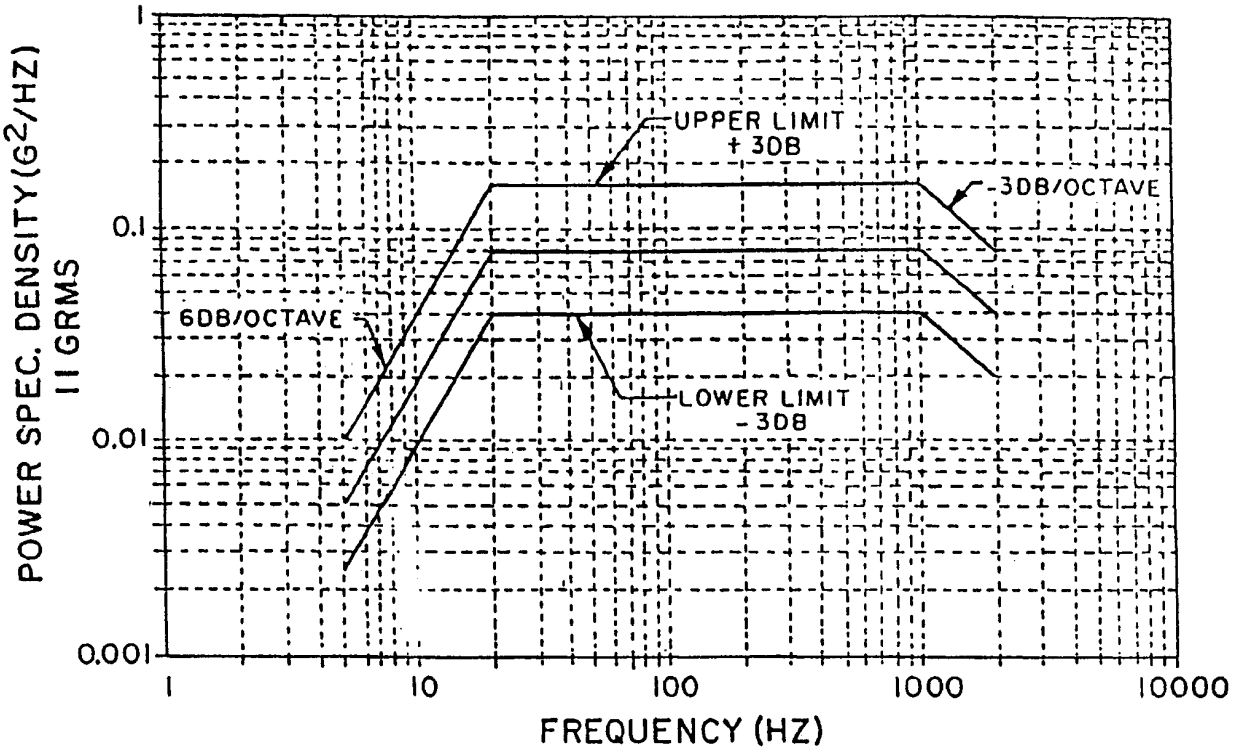
Preferred Marking



Alternate Acceptable Marking

Where marking area is limited by base size, length or size of marking, or where pre-marked bases are used; the letter “Q” may be marked on any convenient space on the base.

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Freq Hz	Vibration Level PSD (G^2/Hz)	Upper Limit (+3 DB) PSD (G^2/Hz)	Lower Limit (-3 DB) PSD (G^2/Hz)
5	0.005	0.01	0.0025
10	0.02	0.04	0.01
20	0.08	0.16	0.04
1000	0.08	0.16	0.04
2000	0.04	0.08	0.02

FIGURE 2. Random vibration curve for T-1-3/4 lamps.

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15. Interchangeability: These lamps may be similar to, but are not interchangeable with, other commercial or military lamps due to unique aircraft applications and qualification.

There are no technical changes in this revision.

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99

Review activities:
Navy - MC, SH
Air Force - 11, 82
DLA - GS

Preparing activity
Navy - AS
(Project 6240-1441-08)

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